NeuroFIDELITY: A Brain Twin Discovery Framework for Alzheimer's Disease

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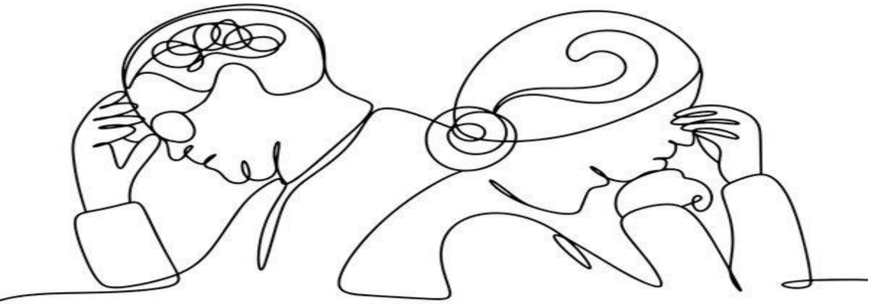
02/06/2026

What is Alzheimer's Disease?

Risk Factors

Symptoms

Diagnosis & Treatment



Literature, Relevance, and Motivation

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- Disease Classification
- Population Level Analysis
- Limited Personalization

Objectives

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01 Extract meaningful MRI representations from Alzheimer's disease scans

02 Encode patient representations into a quantum state space

03 Retrieve and analyze clinically similar patients using quantum fidelity

Dataset(s)

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OASIS-1 MRI Dataset

- 461 subjects
- ~80,000 MRI slices
- 256 × 256 image resolution

- Final Dataset
  - 1,000 MRI images
  - 4 Alzheimer's disease stages
    - Non Demented
    - Very Mild Demented
    - Mild Demented
    - Moderate Demented

Methodology & Workflow

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- MRI Dataset
- CNN Feature Extraction
- PCA Dimensionality Decrease
- Classical Similarity Check
- Quantum Similarity Check
- Perform Comparison

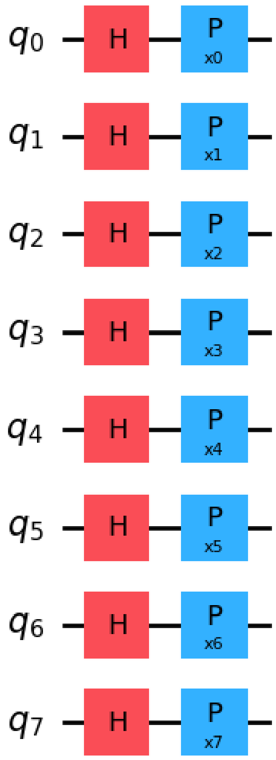
Classical Similarity Calculation

Method		
	Cosine Similarity	Angular similarity
	Euclidean Distance	Distance between vectors
	Manhattan Distance	Sum of absolute differences

Limitation of Classical Metrics

Category	Cos Similarity	Euclidean Distance	Manhattan Distance
Non-Demented	45.7%	46.5%	45.6%
Very Mild Demented	38.1%	39.8%	40.6%
Mild Demented	50.0%	52.5%	50.8%
Moderate Demented	98.2%	98.8%	98.6%

Qiskit Implementation

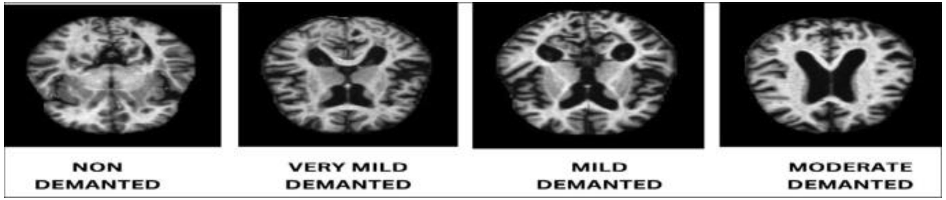


- Phase encoding (vector of parameters $\vec{x}_i$ for each data point)
  - Denoted $\Phi(\vec{x})$
- Obtain similarity by fidelity calculation: $\left|\langle\Phi(\vec{x}_i)|\Phi(\vec{x}_j)\rangle\right|$
  - Use as measure of “closeness of states”
  - Kernel
  - Same results from angle encoding

Results

Brain Twin Retrieval Example

Query ID : 201 (NON DEMENTED PATIENT)

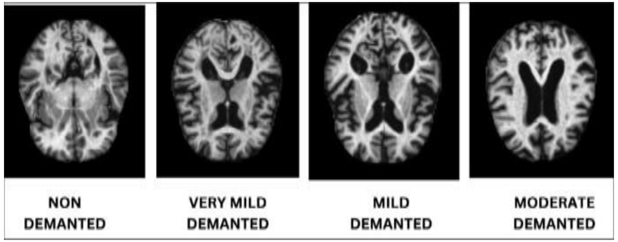


Method	Retrieved Neighbors
Quantum Fidelity	11 ND + 9 VMD
Euclidean	9 ND + 11 VMD
Cosine	8 ND + 12 VMD
Manhattan	8 ND + 12 VMD

Results

Brain Twin Retrieval Example

Query ID : 965(MODERATE DEMENTIA)



Method	Retrieved Neighbors
Quantum Fidelity	19 Moderate Dementia + 1 Mild Dementia
Euclidean	17 Moderate Dementia + 3 Mild Dementia
Cosine	18 Moderate Dementia + 2 Mild Dementia
Manhattan	17 Moderate Dementia + 3 Mild Dementia

Key Results and Conclusion

- Developed NeuroFIDELITY
- Quantum shows promise
  - Able to take advantage of high-dimensional space for analysis of parameters, reduced to 8-dimensional space
  - Highest similarity for moderate demented patients
  - First two categories, tends to perform better than classical alternatives
  - Retrieved **19/20 same-stage brain twins** for a Moderate Demented patient
  - Successfully identified clinically similar patients from structural MRI scans.
  - Future steps: full quantum kernel machine learning, addition of extra parameters

Thank you!!!

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